

Robust Station Selection for Microtransit Under Demand Uncertainty

Gabriel Yong

1 Introduction

Microtransit is a form of transit that seeks to fill the gap between existing services such as private-hire vehicles and scheduled public transit. It is an on-demand, publicly accessible shared-ride service. Like private-hire, it is flexible and on-demand, but it also benefits from the economies of scale of scheduled public transportation by using mid-size capacity vehicles and allowing for greater vehicle utilization.

However, having a fully flexible microtransit system is overly complex and reduces its aggregation capabilities, often having to make significant detours just to satisfy one or two customers. Thus, selection of these *virtual bus stops* (VBS) becomes an important strategic decision. The optimal selection enables better passenger aggregation, reducing total vehicle distance travelled and improving vehicle utilization, while offering passengers decreased waiting times and fewer detours than scheduled public transit.

While the VBS selection is able to adapt between service periods within the day (morning, afternoon, evening, night), the selection for each service period should be robust against shifts in the demand patterns. Within a service period, demand can fluctuate from day to day and the system should still provide a relatively similar experience across each day. We thus address this problem with a two-stage robust model where origin-destination (OD) demand varies within a budgeted uncertainty set.

2 Model

We formulate VBS selection as a two-stage mixed-integer program. The first stage designates a fixed set of l stations shared across all service periods. The second stage activates k of these per period, and OD pair assignments are used to evaluate the demand-weighted walking and routing cost of each activation plan. We first present the nominal model assuming known demand, then characterise the demand uncertainty set, and finally derive the robust counterpart.

2.1 Nominal Model

Sets.

- J set of candidate stations,
- S set of service scenarios,
- Ω set of origin-destination pairs,
- A_{od} admissible pickup-dropoff station pairs for OD pair (o, d) ,

where

$$A_{od} = \left\{ (j, k) \in J \times J : d_{oj}^{\text{orig}} \leq W^{\text{max}}, d_{dk}^{\text{dest}} \leq W^{\text{max}} \right\}.$$

In the experimental instance, A_{od} is non-empty for all $(o, d) \in \Omega$.

Input parameters. The following parameters are assumed to be known:

$q_{ods} \geq 0$	demand for OD pair (o, d) in scenario s ,
$d_{oj}^{\text{orig}} \geq 0$	walking distance from origin o to station j ,
$d_{dk}^{\text{dest}} \geq 0$	walking distance from station k to destination d ,
$c_{jk} \geq 0$	vehicle travel cost from station j to station k ,
$\lambda \geq 0$	weight on vehicle travel cost relative to walking distance,
$W^{\text{max}} \geq 0$	maximum walking distance allowed for a passenger,
$l \in \mathbb{Z}_+$	number of stations to select,
$k \in \mathbb{Z}_+$	number of stations to activate per scenario.

Decision variables. The model has three layers of binary decisions:

$$\begin{aligned}
y_j &= \begin{cases} 1 & \text{station } j \text{ is selected} \\ 0 & \text{otherwise} \end{cases} \\
z_{js} &= \begin{cases} 1 & \text{station } j \text{ is activated and available for service in scenario } s \\ 0 & \text{otherwise} \end{cases} \\
x_{odjks} &= \begin{cases} 1 & \text{OD pair } (o, d) \text{ is assigned to pickup station } j \text{ and dropoff station } k \text{ in scenario } s \\ 0 & \text{otherwise} \end{cases}
\end{aligned}$$

Objective. The objective minimizes total demand-weighted travel cost across all scenarios:

$$\min_{x, y, z} \sum_{s \in S} \sum_{(o, d) \in \Omega} \sum_{(j, k) \in A_{od}} q_{ods} a_{odjks} x_{odjks} \quad (1)$$

$$\text{where } a_{odjks} = d_{oj}^{\text{orig}} + d_{dk}^{\text{dest}} + \lambda c_{jk}$$

Constraints. The constraints enforce the two-stage structure:

$$\sum_{(j, k) \in A_{od}} x_{odjks} = 1 \quad \forall (o, d), s \quad (2)$$

$$x_{odjks} \leq z_{js}, \quad x_{odjks} \leq z_{ks} \quad \forall (o, d), (j, k) \in A_{od}, s \quad (3)$$

$$z_{js} \leq y_j \quad \forall j, s \quad (4)$$

$$\sum_j y_j = l, \quad \sum_j z_{js} = k \quad \forall s \quad (5)$$

Constraint (2) requires every OD pair to be assigned to exactly one admissible station pair per scenario. Constraints (3) ensure that assignments can only use active stations. Constraint (4) requires that a station must be selected before it can be activated. Constraint (5) fixes the number of selected stations to l and the number of active stations per scenario to k .

2.2 Robust Counterpart

The nominal model assumes demand q_{ods} is known exactly, but in practice it varies from day to day. To guard against this, we require the station selection to perform well under any demand realisation within a budget uncertainty set.

2.2.1 Demand Uncertainty

For each period s , demand lies in

$$\mathcal{U}_s = \left\{ q_s : \underline{q}_{ods} \leq q_{ods} \leq \bar{q}_{ods} \ \forall (o, d), \sum_{(o,d) \in \Omega} q_{ods} \leq \bar{Q}_s, q_{ods} \in \mathbb{Z}_+ \forall (o, d) \right\}. \quad (6)$$

The per-OD bounds $[\underline{q}_{ods}, \bar{q}_{ods}]$ reflect the observation that demand for a given OD pair fluctuates around a historical range from day to day. There is also an aggregate limit defined by $\bar{Q}_s$, which represents the idea that while demand can shift across OD pairs, the total number of trips within a time period is unlikely to increase beyond the population limit. Together, this uncertainty set is able to accurately represent worst case demand shifts over each scenario.

One concern is that since passenger demand is integral, this uncertainty set might be difficult to optimize over, but since the bounds $\underline{q}_{ods}, \bar{q}_{ods}, \bar{Q}_s$ should be integer values, the extreme points of the linear relaxation of the uncertainty set will also be integer. Thus, integrality can be dropped from the uncertainty set without changing the worst-case value.

2.2.2 Robust Objective and Recourse

Given fixed activation decisions z_s and realized demand q_s , the optimal assignment cost is captured by the recourse value:

$$\Phi_s(z_s, q_s) := \min_{x \in \{0,1\}} \sum_{(o,d) \in \Omega} q_{ods} \left(\sum_{(j,k) \in A_{od}} a_{odjks} x_{odjks} \right) \quad (7)$$

subject to (2)–(3). Since this inner problem decomposes by OD pair, letting

$$m_{ods}(z_s) = \min_{\substack{(j,k) \in A_{od} \\ z_{js}=1, z_{ks}=1}} a_{odjks}$$

denote the cheapest admissible station pair where both pickup and dropoff are active in scenario s , the recourse value simplifies to

$$\Phi_s(z_s, q_s) = \sum_{(o,d) \in \Omega} q_{ods} m_{ods}(z_s). \quad (8)$$

The robust objective then minimizes the worst-case total cost across all scenarios:

$$\min_{y,z} \sum_{s \in S} \max_{q_s \in \mathcal{U}_s} \Phi_s(z_s, q_s), \quad (9)$$

subject to (4)–(5). As (8) shows, Φ_s is linear in q_s for fixed z_s , so the inner maximisation over $q_s \in \mathcal{U}_s$ is a linear program in q_s and can be dualized to yield a tractable reformulation.

2.2.3 Derivation

Introducing epigraph variables η_s , the robust problem becomes

$$\min_{y,z,\eta} \sum_s \eta_s \quad (10)$$

subject to

$$\begin{aligned} \eta_s &\geq \max_{q_s \in \mathcal{U}_s} \Phi_s(z_s, q_s) && \forall s, \\ z_{js} &\leq y_j && \forall j, s, \\ \sum_j y_j &= l, \quad \sum_j z_{js} = k && \forall s, \\ y_j, z_{js} &\in \{0, 1\}. \end{aligned}$$

Substituting (8) and dualizing the inner problem gives dual variables $\alpha_s \geq 0$, $\beta_{ods} \geq 0$, $\gamma_{ods} \geq 0$ such that

$$\eta_s \geq \bar{Q}_s \alpha_s + \sum_{(o,d)} \bar{q}_{ods} \beta_{ods} - \sum_{(o,d)} \underline{q}_{ods} \gamma_{ods} \quad \forall s, \quad (11)$$

$$\alpha_s + \beta_{ods} - \gamma_{ods} \geq m_{ods}(z_s) \quad \forall (o, d), s. \quad (12)$$

Expanding $m_{ods}(z_s)$ with the assignment variables x and eliminating η_s yields the final MILP:

$$\min_{x,y,z,\alpha,\beta,\gamma} \sum_s \bar{Q}_s \alpha_s + \sum_s \sum_{(o,d)} \bar{q}_{ods} \beta_{ods} - \sum_s \sum_{(o,d)} \underline{q}_{ods} \gamma_{ods} \quad (13)$$

subject to

$$\alpha_s + \beta_{ods} - \gamma_{ods} \geq \sum_{(j,k) \in A_{od}} a_{odjks} x_{odjks} \quad \forall (o, d), s, \quad (14)$$

$$\sum_{(j,k) \in A_{od}} x_{odjks} = 1 \quad \forall (o, d), s, \quad (15)$$

$$x_{odjks} \leq z_{js}, \quad x_{odjks} \leq z_{ks} \quad \forall (o, d), (j, k) \in A_{od}, s, \quad (16)$$

$$z_{js} \leq y_j \quad \forall j, s, \quad (17)$$

$$\sum_j y_j = l, \quad \sum_j z_{js} = k \quad \forall s, \quad (18)$$

with $\alpha_s, \beta_{ods}, \gamma_{ods} \geq 0$ and $x_{odjks}, y_j, z_{js} \in \{0, 1\}$.

3 Experiments

We use operational data from a microtransit system in Zhuzhou, China. The full network has 84 stations, but due to scalability issues we reduce to the 40 most popular stations by OD request volume. Because these stations account for the majority of demand, the reduction in order coverage is minimal.

Table 1 describes the three variants. The sparse variant is constructed by randomly reassigning demand counts from the least popular OD pairs onto the most popular ones, concentrating all demand onto the top $\sim 14\%$ of OD pairs ($956 \rightarrow 131$). This mimics a data-poor setting where granular OD-level observations are less available.

Table 1: Dataset variants for the April and May demand windows.

Metric	Full Zhuzhou	Sampled	Sparse
Stations	84	40	40
April orders	25,947	24,476	24,476
April ODs with orders	1,426	956	131
May orders	31,546	30,100	30,100
May ODs with orders	1,412	949	131

3.1 Experimental Design

We optimize station selection over data on April 1–30, 2025 and evaluate on May 1–31, 2025. We fix $l = 30$ selected stations, a 300-second walking limit, $\lambda = 10$, and vary $k \in \{10, 15, 20\}$. For the robust model we vary the calibration quantile $\tau \in \{0.90, 0.95, 0.99\}$.

3.2 Nominal Feasible Model

As a baseline that enforces service coverage, we introduce the *nominal feasible model*. It extends the nominal model with per-station coverage constraints:

$$\sum_{k \in N_j} z_{k,s} \geq 1 \quad \forall j \in J, s \in S,$$

where $N_j = \{k \in J : d_{jk}^{\text{walk}} \leq W^{\max}\}$ is the set of stations reachable from location j on foot, so the constraint requires that at least one station within walking distance of every candidate location is activated in each scenario. This guarantees that all possible demand locations are serviceable. Unlike the robust model, the nominal feasible model uses the same averaged demand in its objective as the nominal model.

3.3 Uncertainty-Set Calibration

Scenarios correspond to four time-of-day periods: morning (06–10h), afternoon (10–15h), evening (15–20h), and night (20–24h).

Lower bounds $\underline{q}_{ods}$: $\underline{q}_{ods} = 0$ for all OD pairs, since many pairs are absent on any given day.

Upper bounds $\bar{q}_{ods}$: Every OD pair in the full network receives a strictly positive upper bound $\bar{q}_{ods}$. For pairs observed at least once in the calibration window, $\bar{q}_{ods}$ is the empirical τ -quantile of daily counts. For pairs with no observed demand, or pairs whose quantile rounds to zero, $\bar{q}_{ods}$ is set to a flat floor equal to the minimum positive $\bar{q}$ observed across all OD pairs and periods.

Aggregate cap $\bar{Q}_s$: $\bar{Q}_s$ is the τ -quantile of total period demand across calibration days.

Table 2 shows that the budget constraint is meaningful. Without it, the adversary could simultaneously push every OD pair to its upper bound, inflating total demand.

Table 2: Uncertainty-set diagnostics, 90th-percentile calibration.

Period	Sampled ODs	Sparse ODs	Sampled ratio	Sparse ratio
Morning (06–10h)	341	129	1.54	1.83
Afternoon (10–15h)	612	131	1.41	1.34
Evening (15–20h)	725	131	1.44	1.34
Night (20–24h)	351	128	1.50	1.59

4 Results

Experiments were run on the downsampled 40 station dataset and the models were optimized with data from April and tested on the full May period. In the sparse case, models were optimized using demand data from 131 ODs while the evaluation on the May data spans a wider set of ODs.

4.1 Sampled-Network Results

With dense OD coverage in the full sample, all models achieve zero uncovered orders (Table 3). The nominal and nominal feasible models achieve the lowest mean cost. Robust variants have a slightly higher average cost but generally have lower per-order cost variance.

It should be expected that the cost between nominal and the nominal feasible model is the same in this case. However they have negligible differences, likely due to the MIP gap in the solver being set to 0.1%.

Table 3: Sampled-network results, full-OD May orders. Bold = best in column within each k -block.

k	τ	Model	April mean cost	May mean cost	April std.	May std.	Uncovered
10	–	Nominal	4262.40	4319.94	1144.72	1098.14	0
10	–	Nominal (feas.)	4261.24	4324.87	1161.92	1110.97	0
10	0.90	Robust	4288.12	4345.74	1154.60	1099.07	0
10	0.95	Robust	4299.23	4369.80	1144.33	1090.94	0
10	0.99	Robust	4299.02	4363.20	1140.96	1088.16	0
15	–	Nominal	3955.90	4028.27	1133.51	1088.94	0
15	–	Nominal (feas.)	3955.90	4028.27	1133.51	1088.94	0
15	0.90	Robust	4020.73	4092.75	1106.56	1052.13	0
15	0.95	Robust	4010.86	4079.16	1093.28	1040.93	0
15	0.99	Robust	4009.53	4086.36	1103.79	1049.66	0
20	–	Nominal	3882.29	3957.45	1165.42	1114.69	0
20	–	Nominal (feas.)	3881.77	3957.47	1167.29	1116.31	0
20	0.90	Robust	3921.15	3999.72	1126.05	1075.51	0
20	0.95	Robust	3926.23	4003.83	1123.39	1071.72	0
20	0.99	Robust	3936.01	4014.41	1123.40	1072.74	0

4.2 Sparse-Network Results

With sparse OD data, the nominal model has the best mean cost but leaves 214 orders uncovered at $k = 10$. The feasible-nominal model achieves complete coverage at $k = 10$ with a lower mean cost than any robust variant (Table 4), demonstrating that explicit coverage constraints can match the

service-availability guarantee of robustness without having to pay the cost premium from protecting against the worst case scenario. At $k = 15$ and $k = 20$, all models cover every order and costs converge.

Table 4: Sparse-network results, full-OD May orders. Bold = best in column within each k -block. Mean costs for Nominal at $k = 10$ exclude 214 uncovered orders.

k	τ	Model	April mean cost	May mean cost	April std.	May std.	Uncovered
10	–	Nominal	4271.53	4324.72	1122.67	1082.10	214
10	–	Nominal (feas.)	4259.78	4318.17	1150.03	1102.56	0
10	0.90	Robust	4295.37	4351.95	1147.97	1092.40	0
10	0.95	Robust	4290.17	4357.09	1148.56	1099.16	0
10	0.99	Robust	4303.98	4376.77	1144.23	1089.74	0
15	–	Nominal	3958.48	4026.29	1117.83	1071.99	0
15	–	Nominal (feas.)	3962.03	4033.63	1122.01	1080.06	0
15	0.90	Robust	4010.00	4074.93	1100.96	1049.98	0
15	0.95	Robust	4020.72	4094.39	1097.51	1042.90	0
15	0.99	Robust	4007.22	4078.84	1103.19	1049.15	0
20	–	Nominal	3886.85	3960.08	1154.76	1105.72	0
20	–	Nominal (feas.)	3889.32	3963.05	1157.47	1107.31	0
20	0.90	Robust	3912.60	3983.97	1126.73	1081.03	0
20	0.95	Robust	3929.62	4006.93	1124.91	1074.98	0
20	0.99	Robust	3938.17	4020.94	1129.85	1077.70	0

Figure 1 shows the activations of each model in the morning period at $k = 10$. The nominal model focuses its activations on serving demand efficiently compared to the nominal feasible model and the robust model, which results in some distant stations not being covered. The nominal feasible model enforces coverage constraints and activates peripheral stations, achieving zero uncovered orders at lower cost than any robust variant. The robust ($\tau = 0.95$) model similarly distributes activations broadly and achieves full coverage but at a higher cost than the nominal feasible model.

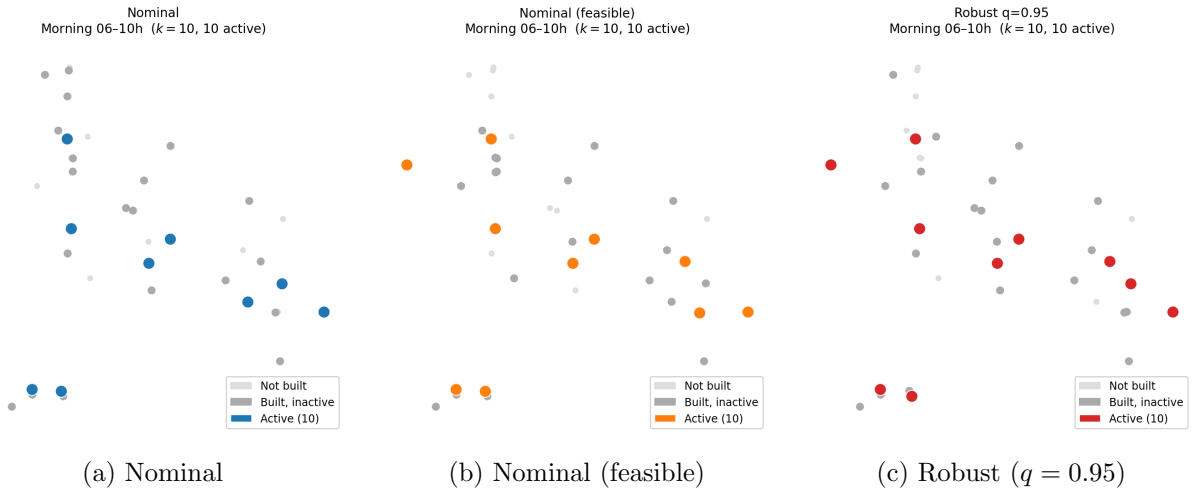


Figure 1: Morning-period activations for $k = 10$, sparse experiment. Filled = active; hollow = selected but inactive.

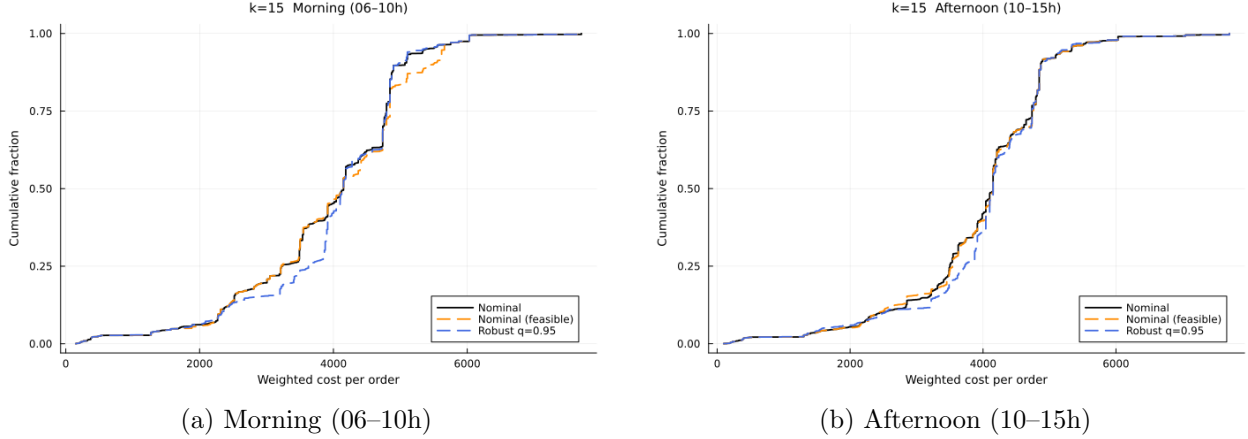


Figure 2: Cost CDFs for $k = 15$, sparse experiment, May full-OD test set.

4.3 Understanding the Lower Spread in Cost for Robust Model

In Figure 2, we see that the lower standard deviation of the robust model is not due to the reduction in cost of the orders that experience high cost in the nominal model, but rather due to the increase in cost for orders that experience low cost in the nominal model. This shows that although the solution is robust, it does not make any improvement over the nominal solution, and provides a strictly worse solution from a cost perspective.

This is likely due to a geometric limitation of the problem. The limited station pool may not provide sufficient spatial variety for the robust solution to reduce the cost of orders that experience high cost under the nominal solution.

5 Conclusion

A budget uncertainty set is a practical way to model uncertain OD demand in the VBS selection problem. We observe that when data is sparse, a nominal solution may lack service coverage while the robust solution provides service guarantees. However, the robust solution incurs a cost premium without reducing high-end costs, even though the variance in order costs has decreased. Enforcing coverage via explicit constraints is more efficient in this setting, as the priority in transportation is ensuring that the cost experienced by passengers is proportionate to their travel distance, rather than minimising cost variance.

Future work could incorporate richer recourse functions reflecting true microtransit operations (vehicle capacity, rebalancing, multi-period dynamics), and identify candidate station sets and demand patterns that benefit most from a robust solution.

A Additional Figures

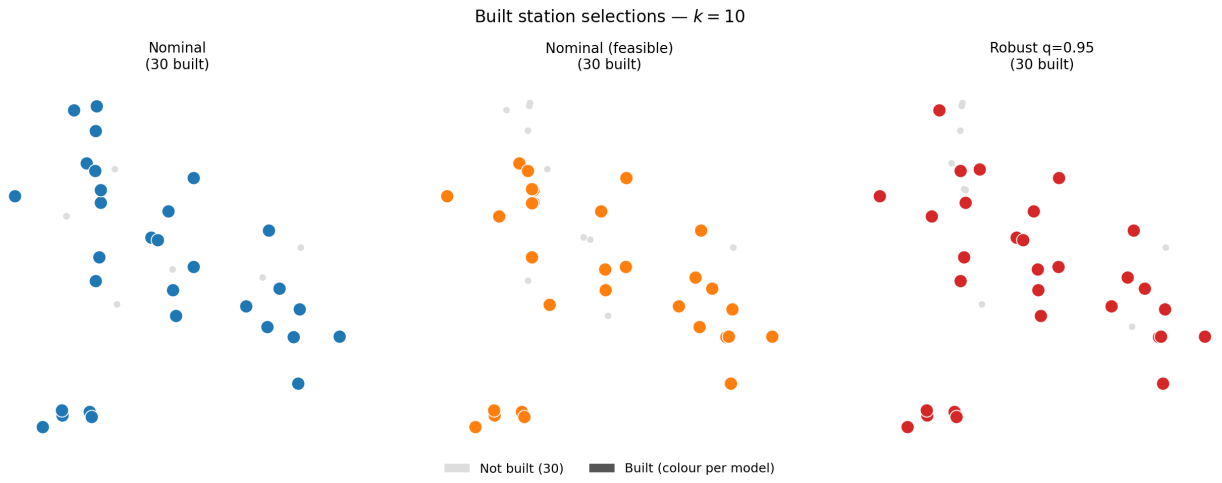


Figure 3: Selected stations for $k = 10$. Coloured dots = selected (30 of 40); gray = not selected. All three models select the same set of 30 stations.

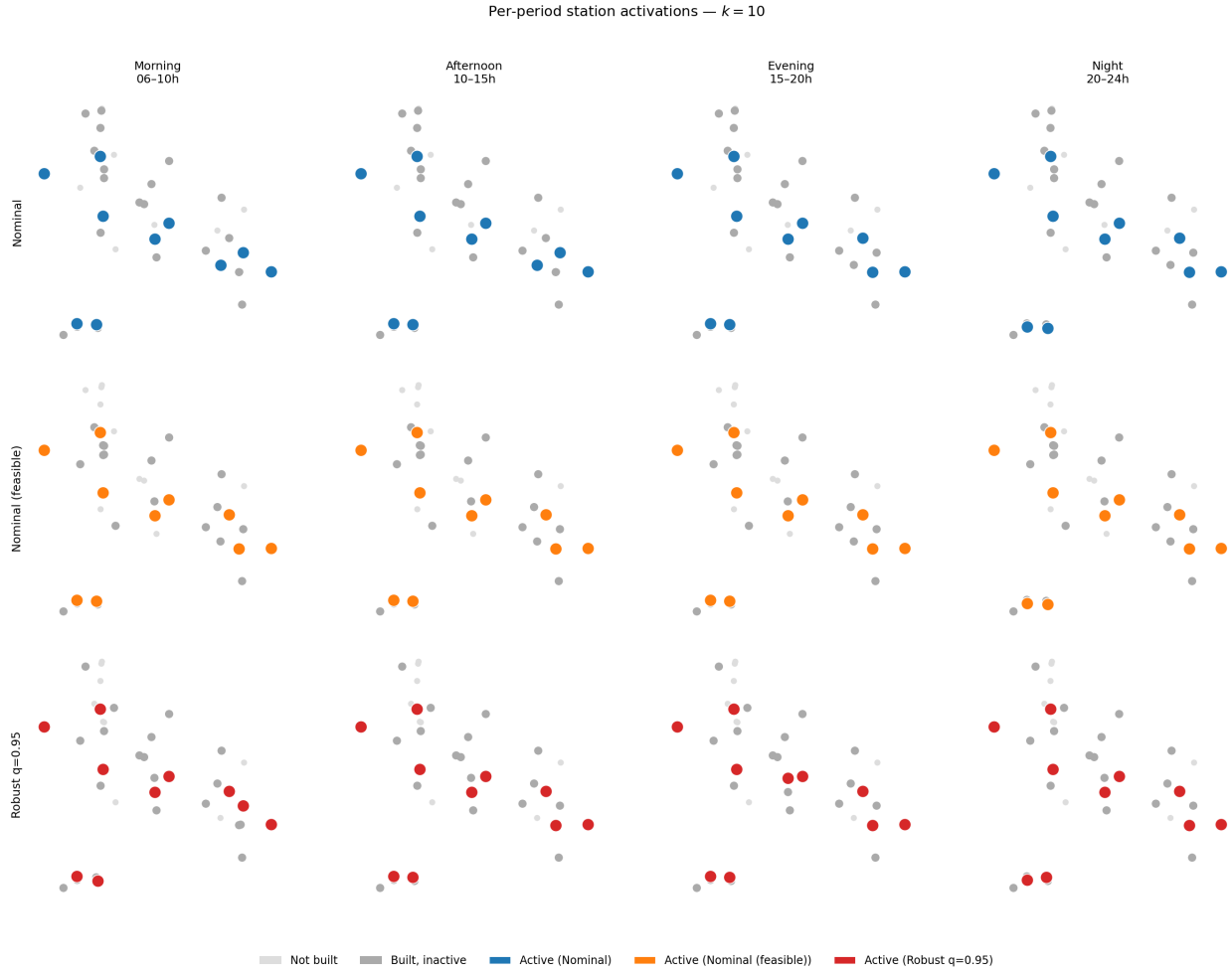
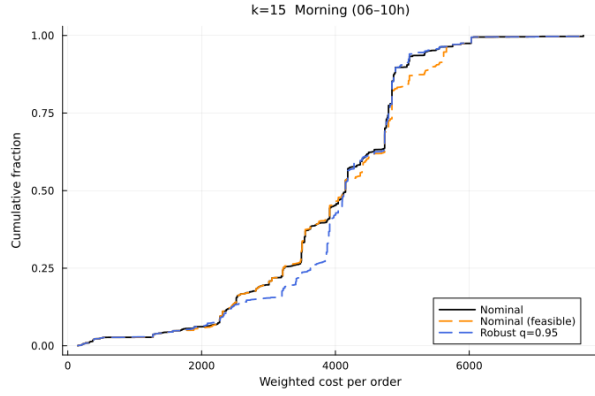
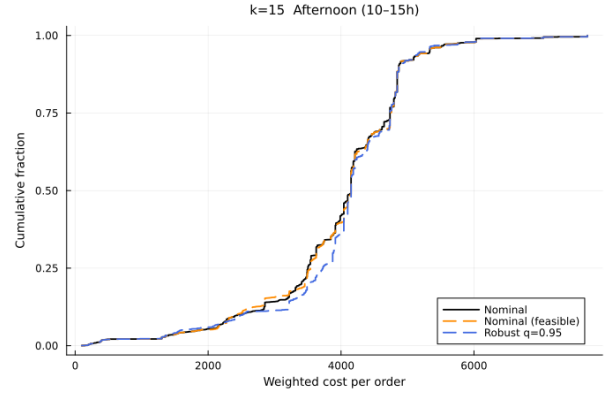


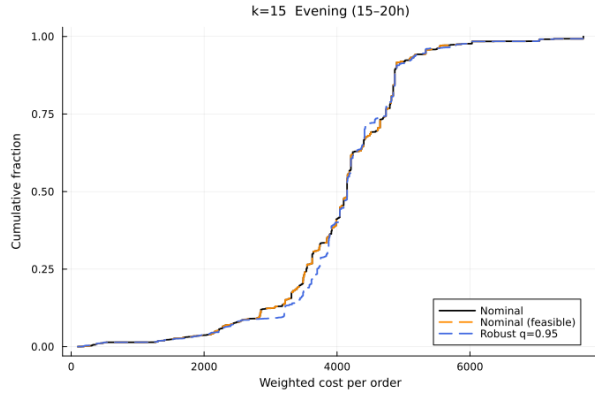
Figure 4: Per-period activations for $k = 10$ across all four time-of-day scenarios. Filled = active; hollow = selected but inactive; gray = not selected.



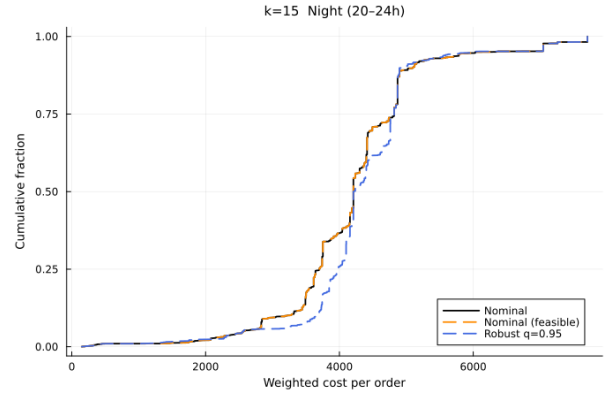
(a) Morning (06-10h)



(b) Afternoon (10-15h)



(c) Evening (15-20h)



(d) Night (20-24h)

Figure 5: Cost CDFs for $k = 15$ across all four time-of-day scenarios, sparse experiment, May full-OD test set.